Ilia Snezhkov

Senior Software Engineer — Backend & Platform · Application Security

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Software engineer with seven years of professional Python across backend and platform work — startups, high-traffic consumer products, regulated fintech and telecom — specialising in application security. Builds the platforms and automation that put security scanning, triage and remediation inside the CI/CD pipelines teams already use. Has built an application-security platform from scratch at two companies, leading the technical side of both.

Technical skills

LANGUAGES Python (primary), SQL, Bash

BACKEND FastAPI, Litestar, gRPC, asyncpg, Celery, Django, Flask

DATA PostgreSQL, Redis, MongoDB, SQLite, RabbitMQ

INFRASTRUCTURE Docker, Kubernetes, service mesh, event-driven architecture, microservices

CI/CD GitLab CI/CD, pipeline automation, quality gates, Prometheus, Grafana, Sentry

SECURITY Secure SDLC, DevSecOps, threat modelling (STRIDE), secure code review, OWASP Top 10, vulnerability management and triage

SECURITY TOOLING SCA/SBOM, SAST, secrets detection, IaC scanning; Semgrep, Trivy, Syft, gitleaks, KICS

AI-ASSISTED DELIVERY LLMs in development and security workflows — false-positive triage, automated remediation

Experience

Senior Software Engineer — MTS Web Services

Apr 2024 – Present · Russia · Remote

COMPANY-WIDE APPLICATION SECURITY PLATFORM

The technology arm of MTS, one of Russia's largest telecom groups — public cloud, data, AI and developer platforms for enterprise customers and for the group's own engineering organisation.

An internal DevSecOps platform that collects security data wherever code and workloads live — scanners in GitLab CI, image scanning in Kubernetes clusters, cloud inventory — and drives everything downstream from it: CI quality gates, auto-fix merge requests, and the security team's own back-office and analytics.

- Own the platform end to end — architecture, backend, data model, automated remediation, and both the security-team and developer-facing surfaces. Architecture and implementation decisions are mine to make, within a direction agreed with engineering leadership.
- Built for scale: thousands of repositories across dozens of engineering teams and clusters, tens of thousands of container images and releases analysed, and a dependency inventory of hundreds of thousands of components.
- Built CVE prioritisation over a knowledge base of published vulnerabilities enriched with exploitation signals (CISA KEV, public exploits, EPSS), so triage starts from what is actually being exploited rather than from raw scanner volume.
- Built CVE-exposure analysis: when a new vulnerability is disclosed, affected projects surface automatically and reach the security team as tracked work, without waiting on input from product teams.
- Integrated multiple classes of scanning (SCA/SBOM, SAST, secrets, IaC) into CI, combining open-source tools with in-house extensions and custom scanners.
- Introduced LLMs as an automated second opinion on scanner findings — cutting false positives and a large share of the team's manual triage load.

- Exposed the platform over MCP — one interface onto the same inventory, findings and remediation, used by people and agents alike. On it, agents that assist the security team's operators through triage, secure code review and penetration testing, with the judgement left to them.

PYTHON · FASTAPI · POSTGRESQL · KUBERNETES · DOCKER · GITLAB CI/CD · MCP · SEMGREP · TRIVY · SYFT · GITLEAKS · TYPESCRIPT

Senior Software Engineer — Raiffeisen Bank Russia

May 2022 – Apr 2024 · Russia · Remote

BANK-WIDE APPLICATION SECURITY PLATFORM

Russian subsidiary of Raiffeisen Bank International — retail and corporate banking in a heavily regulated environment.

Led the build of the platform from scratch: security scanning, triage and reporting embedded across the SDLC of a large, predominantly Java organisation under strict audit and compliance requirements.

- Security checks were built to fit the delivery flows teams already had, rather than requiring them to standardise their pipelines first.
- Built and maintained the platform's async Python microservices and contributed to its React/TypeScript frontend.
- Automated finding triage and reporting so results reached owning teams as actionable work rather than raw scanner output.
- Owned CI/CD and observability end to end — Kubernetes, GitLab pipelines, Grafana dashboards, error tracing.

PYTHON · FASTAPI · LITESTAR · ASYNCPG · POSTGRESQL · KUBERNETES · GITLAB CI/CD · REACT · GRAFANA

Senior Python Engineer — Master Delivery

Oct 2021 – May 2022 · Russia · Remote

Last-mile delivery provider carrying merchants' online orders through to delivery.

Backend microservices and shared internal libraries used across the company's engineering teams.

- Developed FastAPI microservices and reusable internal libraries adopted by other backend teams.
- Led refactoring and performance optimisation of existing services.

PYTHON · FASTAPI · POSTGRESQL · DOCKER

Senior Software Engineer — McDonald's Russia

Apr 2020 – Sep 2021 · Russia · Remote

MOBILE APP & WEBSITE BACKEND API

Backend for the McDonald's Russia mobile app and website, built under contract via ADV — around 15 million users, one of the country's highest-traffic consumer products.

A high-load API for the mobile app and the website — gRPC microservices behind a service mesh, event sourcing for state changes, and product features through to the payment integration. Peaks near 10,000 requests a second were absorbed by optimisation rather than by more hardware.

- Shipped product features end to end, including the payment integration.
- Optimised the services carrying the peak so the load was met on the existing cluster rather than by scaling the infrastructure.
- Introduced a service mesh and event sourcing into the platform architecture, improving service-to-service reliability and auditability of state changes.
- Built gRPC microservices and an internal integration-test framework used across the platform.

PYTHON · GRPC · POSTGRESQL · REDIS · RABBITMQ · DOCKER · KUBERNETES

Python Developer — SQUILLA

Jun 2019 – Apr 2020 · Krasnodar, Russia

Crypto-loans platform — a security-sensitive financial product allowing users to borrow against cryptocurrency collateral.

An API gateway fronting the platform's services, plus a real-time collector aggregating cryptocurrency quotes from multiple exchanges to price collateral.

- Built both services: the gateway unifying access to the backend, and the quotes collector feeding near real-time pricing into collateral valuation.

- Maintained the core Django services of the platform.

PYTHON · FASTAPI · CELERY · DJANGO · POSTGRESQL · REDIS

Python Developer · Contract — Upwork Oct 2016 – Jun 2019 · Remote

Contract backend engineering for international clients, delivered independently end to end. Included a flight-ticket meta-search service with distributed parsing and real-time result streaming, and a SaaS that prioritised test runs by predicting which commits were likely to introduce defects.

PYTHON · FLASK · SOCKET.IO · CELERY · PYTORCH · SCIKIT-LEARN · POSTGRESQL

Personal projects

Suji

Japanese learning platform for iPadOS, iOS and macOS. Native Swift 6 under strict concurrency across 40+ modules; offline-first local SQLite syncing to Supabase through an outbox queue. FSRS-5 spaced repetition, handwriting graded against KanjiVG with dynamic time warping, on-device and cloud AI, and a full teacher-and-student loop.

SWIFT 6 · SWIFTUI · GRDB/SQLITE · SUPABASE · PENCILKIT · METAL

Gokaku

Offline trainer for Japan's written driving-licence exams, on the App Store. 932 true/false questions, each carrying a verbatim quote from the National Police Agency rulebook and machine-verified against that corpus; 208 official sign images; twelve languages; no account, no network, no data collected.

FLUTTER · DART · KOTLIN/ANDROID · PYTHON

Independent research

MindOne — device support for a MediaTek phone
github.com/iasnezhkov — kernel · device tree

A retail Android phone on a MediaTek Helio G99, sold on Linux 5.10 with no published source for its kernel, device tree or hardware abstraction layers. Hand-ported it to Linux 6.12 by way of 6.1, rebuilding the device tree out of the stock compiled blob and carrying 339 out-of-tree driver trees, 290 of them loading on the running phone. Above the kernel, rewrote closed userspace in the open — an audio HAL, a sensors HAL and four HIDL→AIDL bridges — so the device can be tuned and carried onto later Android releases. Two repositories: the 6.12 kernel current, the 6.1 port archived. It is the phone I carry.

LINUX 6.12 · C · AARCH64 · ANDROID GKI/KMI · DEVICE TREE · GHIDRA · AOSP · LINEAGEOS

Education

Southern Federal University (SFEDU) Sep 2013 – Jun 2018 · Taganrog, Russia

Specialist Degree — 5-year programme, equivalent to Master's level — Information & Computer Systems Security

Teaching assistant, 2016–2018. Competitive CTF player — exploitation, reverse engineering.

Languages

English — C1 · Russian — Native · Japanese — Elementary